Short Adaptive CDIs: Validation and Cross-Linguistic Comparison of Computer-Adaptive Parent Report Measures of Early Vocabulary for Polish and English

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Abstract

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Introduction

MacArthur-Bates Communicative Development Inventories (CDI) are parent-report forms that are used worldwide in research and clinical practice to measure children's language abilities, especially word knowledge. There are over 100 language adaptations of the CDI, including Polish (Smoczyńska et al. 2015), and up to three age versions in each language: Words and Gestures (CDI:WG) for children up to the age of a year and a half, Words and Sentences (CDI:WS) for older children up to around three, and CDI-III for children from three to three and a half or four (exact age ranges vary between adaptations). All CDIs include a vocabulary checklist, a list of words in which parents or other caregivers mark the words that their child "understands" and/or "understands and says." CDIs and similar parental reports draw on caregivers' extensive experiences with the child in a variety of contexts. They demonstrate adequate validity with spontaneous language samples (Gatt et al. 2023; O' Toole and Fletcher 2010) and direct language measures (e.g. Dale 1991; Lemhöfer and Broersma 2011; Smolík and Bytešníková 2021) both in typically-developing children as well as children with an elevated likelihood or diagnosis of ASD (Belteki et al. 2025).

The vocabulary checklists include many words across numerous semantic and lexical categories, varying in their frequency, age-appropriateness, and other properties (e.g., Frank et al. 2021; Garmann et al. 2019). As a result, CDIs tend to include hundreds of words and require a substantial amount of time (e.g., 20-30 minutes) for caregivers to complete. Because of the length, there were many attempts to shorten the CDI word lists. The initial attempts were to simply identify a subset of items from the original lists that correlated most strongly with responses on the full list (Fenson et al. 2000; but later also Ezeizabarrena et al. 2024; Smolík and Bytešníková 2021; Lasorsa et al. 2021; Kim et al. 2014). More recently, Mayor and Mani (2019) administered a random subset of words from the full CDI, combining these with vocabulary data from the WordBank database, and then incorporated information about the child's age, sex, and language status into Bayesian models that predict the child's overall vocabulary size as it would appear on the full CDI. This method provided a reliable estimate of the child's language development level without requiring the complete inventory: using a 25-item lists, CDIs correlated with the full CDIs at $r = .96$, with an SE of .14 and a reliability of .98; using a 50-items CDIs showed slightly lower correlation at $r = .94$ with similar SE of .14 and a reliability of .98. However, as noted by Kachergis et al. (2022), while using age and sex to predict vocabulary improves precision with limited data, these factors can also bias individual assessments by pulling scores toward typical expectations for those demographics.

Other attempts to shorten the CDIs have incorporated Item Response Theory (IRT) and Computerized Adaptive Testing (CAT) (e.g., Makransky et al. 2016; Chai et al. 2020; Kachergis et al. 2022). IRT is a statistical framework used in educational and psychological testing to model the relation between responses to individual test items and the measured ability. Unlike Classical Test Theory (CTT), which focuses on overall test scores in relation to ability, IRT examines each item response and how it contributes to estimating the ability. This ability (often denoted as theta) is represented on a continuum: negative values correspond to lower ability, values near zero to average ability, and positive values to higher ability. A key feature of IRT is its focus on item characteristics (or parameters). In a two-parameter model, each item is described by its difficulty (the ability level required to answer correctly) and discrimination (how well the item distinguishes between individuals of similar ability levels). Once item parameters are known, every response provides information about an individual's ability. In the CDI context, IRT model captures the difficulty and discrimination of vocabulary items and the relation between parental responses to those items and the child's vocabulary knowledge, typically using large datasets such as norming samples. Once an IRT model is established and item parameters are estimated, a computerized adaptive task (CAT) can be constructed. This involves specifying how the task should begin, how the ability is to be estimated, and the criteria for ending the task. In a CAT-CDI, items are presented to the parent one at a time. After each response, the algorithm updates the estimate of the child's vocabulary ability and selects the next item to further refine this estimate. The process continues until either a predetermined number of items is reached or the estimate achieves sufficient precision. IRT and CAT-based tasks are increasingly being developed in many areas of language assessment, including reading (Ma et al. 2025), direct vocabulary testing in older children (Bohn et al. 2024; Bohn et al. 2025) and measuring overall variability in children's learning and development (Frank et al. 2025).

The first IRT-based CAT-CDI was developed by Makransky et al. (2016) who applied IRT models to normative data from the CDI:WS. Using the full CDI dataset, they simulated CAT-CDIs of different lengths by sequentially selecting items and filling in responses based on participants' full CDI data. Then they compared the full CDI scores with the scores from various fixed-length CATs ranging from 5 to 400 items. Notably, a CAT with just 50 items produced scores that correlated strongly with the full CDI, achieving a correlation of $r = .95$. Building on this, Kachergis et al. (2022) developed online CATs for assessing vocabulary in both English and Mexican Spanish and extended the adaptive assessment framework to both receptive and productive vocabulary. Their CAT-CDIs were found to accurately estimate vocabulary comprehension and production using as few as 25–50 items, significantly reducing assessment time while maintaining high validity.

These developments paved the way for the creation of a CAT-CDI in Polish (Krajewski et al., in preparation). While a detailed description of the development process for both CAT-CDIs is beyond the scope of this paper, we briefly highlight key decisions that researchers must consider when developing a CAT-CDI in a new language. To illustrate, we draw on the English and Polish CAT-CDIs as examples. The primary aim of the present paper is to evaluate the performance of these two instruments in validation studies conducted in the USA and Poland. We believe that such an evaluation can guide future CAT-CDI development, enabling researchers to make more informed and strategic decisions.

Item parameters calibration. In both the English (Kachergis et al. 2022) and Polish (Krajewski et al., in preparation) CAT-CDIs, we applied a two-parameter logistic (2PL) model, in which each item is defined by two characteristics: difficulty and discrimination. These parameters were estimated from large datasets: Wordbank data for English and CDI:WS norming data for Polish. A central design decision concerns how many CAT-CDIs to develop and what data to use for item parameters calibration. For English, two instruments were created: one for comprehension and one for production. Unlike the original CDIs, they are not divided into age-based versions. Instead, both CAT-CDIs span the entire developmental range of the CDI instruments, from the earliest CDI:WG through the CDI-III. This design reflects the assumption that early vocabulary can be captured by two latent traits—comprehension and production—and it maximizes the dataset available for parameter estimation. A practical benefit is that practitioners only need to choose between comprehension and production, without worrying about which age-specific version to administer. In contrast, the Polish CAT-CDIs retain the original structure, with separate adaptive instruments for each age version (e.g., CAT-CDI:WS for word production). This structural difference affects how item characteristics are estimated. The English CDI parameters were calibrated on data from children aged 12–36 months (Kachergis et al. 2022), covering the full CDI age range. By comparison, the Polish CAT-CDI:WS was calibrated on children aged 18–36 months, consistent with the Polish CDI:WS norming data. As a result, an item may appear more difficult in English CAT-CDI (relative to Polish) simply because its parameters were estimated using responses from a younger and more heterogeneous sample. At the same time, we can still expect cross-linguistic similarities – both at the item level (e.g., the word "mommy" is likely to be easy in both Polish and English) and at the category level (e.g., sound-related words tend to be generally easier across languages).

CAT settings. In both the English and Polish CAT-CDIs, basic caregiver-reported information – specifically the child's age and sex – is used to generate an initial ability estimate. The first item is then selected based on this estimate. In both versions, we deliberately present a slightly easier item (below the estimated ability level) to encourage a positive response to the first item. Parental response to one item at a time allows the CAT algorithm to update the ability estimate and calculate a standard error (SE) of that estimate. Another important decision concerns the stopping criteria of the CAT task. These criteria can be based on a maximum number of administered items or reaching a target level of estimate precision – typically expressed as a standard error (SE) of the estimate. The English CAT-CDI prioritized a shorter CAT typically ending after 50 items, or sooner if a standard error (SE) of 0.15 was reached. In contrast, the Polish CAT-CDI prioritised greater precision, continuing to a maximum of 75 items until an SE of 0.1 was reached. If this level of precision could not be reached, the test concluded after a maximum of 75 items. These differences in CAT settings may result in relatively shorter English CAT-CDIs but relatively more precise estimates obtained from the Polish CAT-CDIs.

The aim of this paper is to evaluate the potential usefulness of CAT-CDIs for both research and clinical applications, using evidence from the validation studies of the two CAT-CDIs in English and in Polish. For a CAT-CDI to be effective in these contexts – whether for screening, diagnosis, or broader research purposes – it should meet the following criteria:

1. Require less administration time.
2. Produce estimates that align closely with scores obtained from the full CDI.
3. Maintain high measurement precision (i.e., low standard error) across the relevant ability range.

4. Avoid repeated reliance on the same subset of items, which is particularly important in longitudinal or repeated-measures designs.

We will report these outcomes in the two CAT-CDIs, in English and in Polish and discuss similarities and differences in the context of how the development of two CAT-CDIs differed.

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Additionally, we will compare the CAT-CDI item difficulty across Polish and English. Some differences in item parameters are to be expected, given that the IRT calibrations for the two languages were based on slightly different samples. In particular, the English data included younger children, since all age versions of the CDI were collapsed into a single dataset, whereas the Polish calibrations drew on a narrower age range. These sampling differences may influence estimated item parameters, particularly difficulty and discrimination. At the same time, such comparison also provides an opportunity to examine potential cross-linguistic effects. Differences in lexical mapping may emerge between Polish and English, reflecting item-specific factors such as word frequency, morphological complexity, or typical contextual usage. For example, morphologically complex items may be more challenging in Polish than in English, whereas high-frequency words may show comparable difficulty levels across both languages. Identifying the extent to which observed differences arise from methodological factors (i.e., sample composition) versus genuine cross-linguistic variation is an additional goal of this analysis.

Methods

Participants

In the American English validation study, 251 parents were recruited from the online survey vendor Prolific. The sample was ethnically diverse, including White, Black, Latino/Hispanic, Asian, Native American, and other races/ethnicities. The average number of years of education attained by the primary caregiver was 15.85 (SD = 2.23). After excluding participants who provided inconsistent information on their child's age, sex, birth weight, caregiver's ZIP code or year of birth (n = 36), and excluding children with bi- or multilingual background (n = 2) or multiple variations of previous exclusion criteria (n = 9), data from 204 children (103 girls, 50.49%) were included in final analyses. The children were aged between 15 and 36 months (M = 26.21, Mdn = 26, SD = 6.37). The average number of years of education attained by the primary caregiver was 15.82 (SD = 2.27).

In the Polish validation study, 147 parents from an ongoing study were invited via email to take part in the validation study. The sample was White (reflecting the limited ethnic diversity within the Polish population). However, there were 28 children multilingual with Polish (as their home language) and some other (societal) language, living outside of Poland. We considered this group to be too large to exclude from the study, but not large enough to warrant a separate set of analyses. However, we included some comparisons between the Polish monolingual and bilingual groups in Supplementary Material. Next, we excluded participants for whom the time between full CDI and CAT testing was longer than 30 days (n = 31), children who were out of the CDI:WS age range when parents filled in either full or CAT CDI (n = 3). The final dataset included 113 parents (49 girls, 43.36%). The children were aged between 18 and 34 months (M = 24.73, Mdn = 24, SD = 4.47). Parents were asked about their education level and 65 (57.52%) reported having higher education or PhD, 1 (0.88%) reported having unfinished higher education, 11 (9.73%) reported having secondary education and 36 (31.86%) did not report their education level.

Materials

The data presented here was gathered with CAT-CDIs in American English and Polish presented above. The American English CAT-CDI word production includes 679 items and the Polish CAT-CDI version of WS: Words and Sentences includes 666 items. There are 395 words (c.a. 58%) common to both CDI-CAT language versions.

Procedure

Both studies obtained the approval of adequate ethics committees. In both validation studies (in Polish and in English) the parents were asked to fill in both the full CDI and the CAT-CDI online. The American team administered both CDIs within the Web-CDI interface (DeMayo et al. 2021) and the Polish team administered both CDIs within a web app (CDI-Online) built with R Shiny (Krajewski et al. 2020). In the validation of American English CDI-CAT, parents were given both CDIs in one sitting, but the order of the CDI versions was counterbalanced. In the Polish validation study, since the CAT-CDI was added as a final step to an ongoing study, the order of the CDIs was fixed, with the full CDI always first. After 1–30 days (M = 8.31, Mdn = 8, SD = 4.90) parents were asked to fill in the CAT-CDI. In both languages, the view of the CDI differs depending on whether it is a full CDI or a CAT-CDI. In the full CDI (in both languages), each page presents all items in a given semantic category. The smallest categories (with fewest items on a page) are connecting words for English (six items) and modal adverbs in Polish (eight items). The biggest category in both languages are action words (103 items in English and 123 items in Polish). In CAT-CDI in both languages, parents are presented with one word at a time and asked to select one of two answers “Yes” or “No” to indicate whether their child says that word. The word is accompanied by information on the semantic category it belongs to.

Results

Above we proposed that for a CAT-CDI to be effective in the research and clinical contexts, it should meet the following criteria:

1. Require less administration time.
2. Produce estimates that align closely with scores obtained from the full CDI.
3. Maintain high measurement precision (i.e., low standard error) across the relevant ability range.
4. Avoid repeated reliance on the same subset of items, which is particularly important in longitudinal or repeated-measures designs.

Additionally, we will compare the CAT-CDI item difficulty across Polish and English.

In the following section we will explore the two CAT-CDI's performance on each of those criteria. The English data presented here were also included in a brief validation report in Kachergis et al. (2022). In the present paper, however, we examine these metrics in greater detail and place them in relation to the validation results from the Polish CAT-CDI.

Length of administration

One of the key metrics for CAT-CDI's usefulness is whether it can shorten the CDI administration compared to the full CDI (but still obtain a reliable estimate of child's lexical ability). We have found that CAT-CDIs were significantly shorter to administer: the English CAT-CDI median time was 342s (~5.7 minutes) (median full CDI time was) and the Polish CAT-CDI median time was 117s (~1.95 minutes) (median full CDI time was 1240s (~20.67 minutes)). Even though the CAT-CDI in Polish seemed shorter (compared to English CAT-CDI), the number of items administered in the two language versions of CAT-CDI was comparable: English M = 31.42, Mdn = 25 (25–50 items), Polish M =

34.71, *Mdn* = 23 (13–75 items). Notably, as the medians indicate, for half of the participants, the estimate precision was obtained relatively early and the CATs ended after 25 (English) or 23 (Polish) items.

Comparison of estimates

Correlations. In both languages, we found similarly strong correlations between the abilities estimated from CAT-CDI and the full CDI scores (English and Polish: r = .86). The correlations were also strong within individual age groups (see Tables 1 and 2). We also obtained IRT ability estimates from the full CDI using the same 2-PL model. Because these estimates draw on responses to all available items in the full CDI, they incorporate more information about each child's ability, reducing random error and increasing precision. Such ability estimates from the full test version are often considered a baseline for comparisons with ability estimates from the CAT version. We correlated the abilities estimated from the CAT-CDI and abilities estimated from full CDI and found strong correlations in both languages, English and Polish: r = .92. Finally, we also correlated the abilities estimated from the full CDI and the full CDI scores, i.e. two measures obtained from the same CDI version. This way we examine consistency between model-based abilities and traditional scores. Again, we found strong correlations in both languages (English: r = .95, Polish: r = 0.94), confirming that both measures reflect the same underlying ability.

Table 1

English: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	0.95	0.85	0.82	0.83	0.59	0.84	0.86
N	26	22	26	30	28	24	48

Table 2

Polish: Correlations between ability estimated by CAT-CDI and ability estimated from full CDI by children's age.

	[15,18)	[18,21)	[21,24)	[24,27)	[27,30)	[30,33)	[33,36]
r ability CAT vs full CDI	NA	0.8	0.94	0.91	0.89	0.95	NA
N	NA	29	22	16	23	22	1

Discrepancy. Next, we checked for the difference between the abilities as estimated by CAT-CDI and the full CDI. For each participant, we calculated the mean squared error (MSE): we subtracted the CAT-CDI estimate from the full estimate and squared this difference to eliminate negative values. The mean squared error in English was 0.57 (Mdn = 0.35, SD = 0.70), and in Polish it was 0.19 (Mdn = 0.08, SD = 0.45). The MSE was relatively high in English (compared to Polish), indicating a relatively larger discrepancy between CAT-CDI ability estimates and full CDI ability estimates in English. But in both languages, the difference in estimates occurred despite strong correlations between the two measures. This difference between the full and CAT-CDI estimates may be explained by at least two reasons. First, a number of participants with extreme discrepancies between the CAT-CDI and full CDI estimates may be inflating the MSE.

Second, there might be a systematic bias, with one estimate higher than the other.

To explore the first of these possibilities, we identified children for whom the estimates from the CAT-CDI and full CDI showed extreme discrepancies, i.e. the difference between the two estimates was removed by 1.5 SD from the mean. There were 15 such cases (7.35%) in the English dataset and 4 cases (1.96%) in the Polish dataset. We then re-calculated the mean squared error without these cases, which yielded an MSE of 0.44 (Mdn = 0.29, SD = 0.44) in English and 0.12 (Mdn = 0.07, SD = 0.14) in Polish. The updated MSE values were only slightly lower than the original ones, suggesting that extreme discrepancies had only a modest effect on the overall error and cannot fully account for the differences between ability estimates from the full CDI and the CAT-CDI in the two languages.

Next, we examined a possible bias in our two ability estimates. We found that the mean CAT ability estimates were higher than the full CDI ability estimates. That was true both in English, $M_D = 0.55$, 95% CI $[0.48, 0.62]$, $t(203) = 15.11$, $p < .001$ (large effect size: Cohen's $d = 1.06$, 95% CI $[0.89, 1.23]$), and in Polish, $M_D = 0.26$, 95% CI $[0.19, 0.32]$, $t(112) = 7.71$, $p < .001$ (medium effect size: Cohen's $d = 0.73$, 95% CI $[0.52, 0.93]$). A closer look showed that the difference between the full and CAT-CDI estimates depended on the child's ability level (see Figure 1). For low ability, the differences between the estimates were larger, and the English CAT-CDI estimates tended to be higher than the full scores. For the highest ability, the estimates from the two CDIs seemed most aligned. In Polish, the CAT-CDI estimates generally aligned more closely with the full CDI estimates across the whole range of ability. Still, a similar trend has emerged as in English, though weaker: for the lowest ability levels, the CAT-CDI estimates were mostly higher than the full CDI estimates. However, for the highest ability levels, the CAT-CDI estimates were in fact lower than the full CDI estimates.

The question that remains is whether the differences between the two estimates (visible especially in the lower ability, as in Figure 1) come from CAT over-estimating

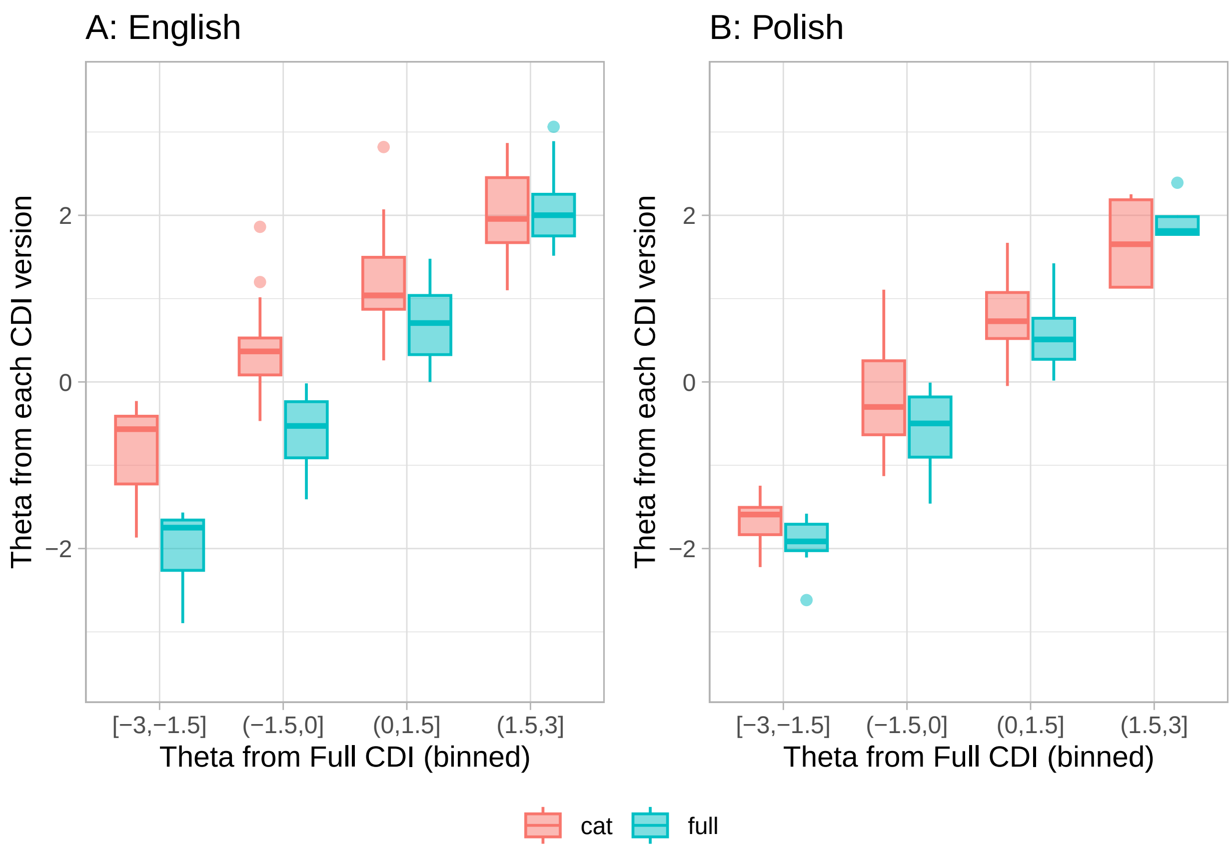


Figure 1. The relation between ability (estimated from full CDI) and the size of discrepancy between the estimates from full CDI and CAT-CDI: (A) English, (B) Polish.

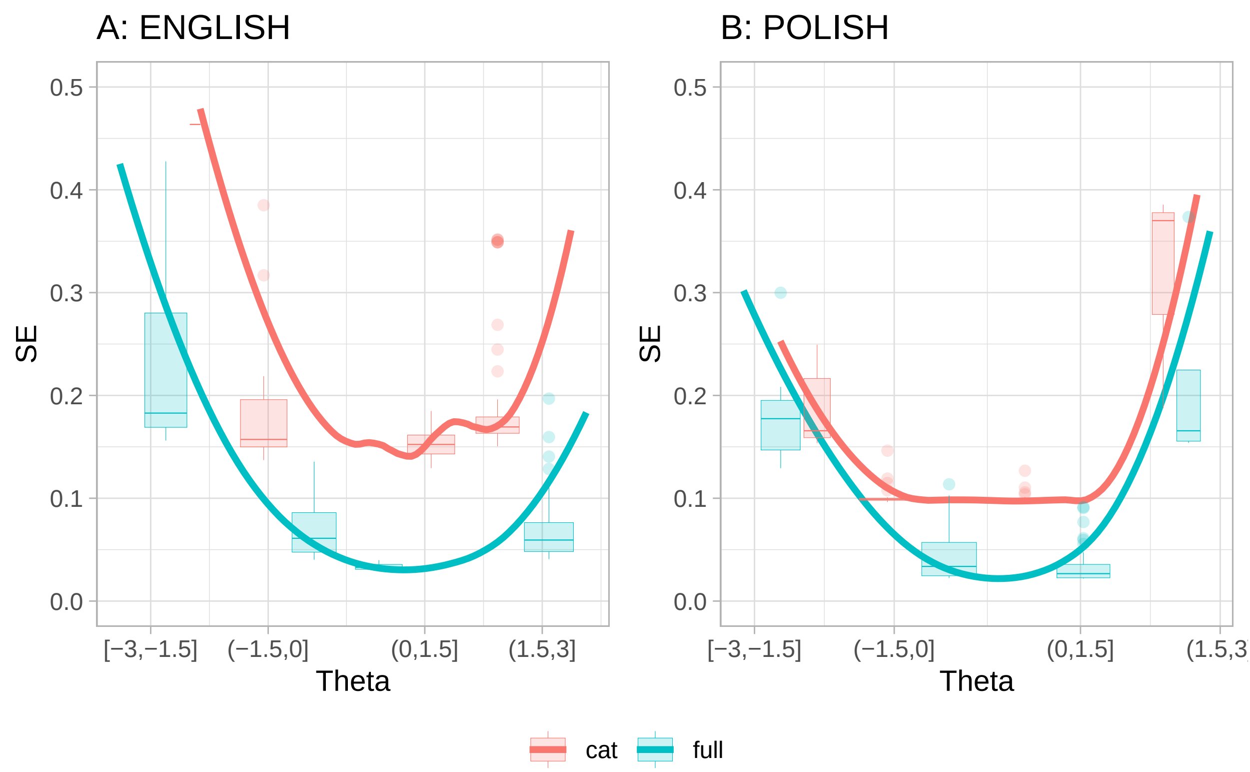
child's true lexical ability or from the full CDI underestimating child's ability. For one, it may be that in CAT-CDI parents answer "yes, my child produces this word" to more items that would be expected (Kachergis et al. 2022), and as a result, CAT-CDI may overestimate the child's true ability. On the other hand, it could be that as the full CDI is very long, parents may omit marking some items that their child produces (see e.g., Łuniewska et al., 2025 for parental inconsistencies in reporting vocabulary of older children). This could result in the full CDI underestimating the child's true ability. It is difficult to answer this question without direct assessment of child's lexical ability. With the existing data, we can at least compare the precision of the two estimates to assess which measure is likely more reliable.

Precision of the estimates

We looked at how precision of the estimate, measured by standard error of the theta (SE), related to the ability levels (see Figure 2). In both languages, the characteristic U-shaped curves indicate that measurement precision is the highest around the middle of the ability range (i.e., showing lowest SE) and decline at the extremes (i.e., for very low or very high ability the SE values are the highest). This U-shape is observed for both the CAT-CDI and the full CDI, indicating that even the full CDI provides relatively less precise estimates at the extreme ends of ability. Note that in Polish (Figure 2, panel B) the lowest ability estimates (i.e., [-3,-1.5]) yielded a slightly lower median SE in CAT-CDI than in the full CDI, suggesting that at lowest ability levels, CAT-CDI might be similarly or slightly more precise than the full CDI. But for the highest ability estimates (i.e., (1.5, 3]) it was the full CDI that came with better precision than the CAT-CDI.

Nevertheless, in both CAT-CDIs, we have been able to reliably estimate lexical ability for the majority of children. According to Makransky, a standard error (SE) below 0.2 is considered acceptable. In practical terms, an SE of 0.2 indicates that the estimated ability is likely within +/- 0.2 units of the child's true lexical ability. Such precision of the CAT-CDI estimate was obtained for 189 out of 204 American participants (92.6%) and 109 out of 113 Polish participants (96.5%). The remaining participants were at the extremes of the ability scale (see Figure 2). The ability estimates from the Polish CAT-CDI generally fell well below an SE of 0.2, except for the highest-ability observations. Because the Polish CAT-CDI was designed to stop once an SE of 0.1 was reached, the precision of Polish estimates never went below 0.1 (i.e., the CAT-CDI was stopped at this point).

Taken together, the results reported so far show that the CAT-CDI estimates correlate strongly with both full CDI scores and estimates obtained from the full CDI. For the lower ability, the CAT-CDI estimates are higher than those of the full CDI, but in the Polish data, the precision of those CAT-CDI estimates is similar or slightly higher than



Note that in English there is only one case where CAT estimate was smaller than −1.5, therefore no CAT boxplot is plotted in that bin.

Figure 2. The relation between ability estimaes from full CDI and CAT-CDI and the standard error of that estimate: (A) English, (B) Polish.

that of the full CDI. This might potentially suggest that for children with lower lexical abilities, CAT estimates may be more aligned with the true abilities. For the higher ability, CAT estimates were more closely aligned with the full CDI estimates, but the precision of the CAT estimates was similar or lower than that of the full CDI. Importantly, for the large majority of children, the obtained CAT-CDI estimates were below the precision cut-off point of 0.2.

Of note here is also the fact that the two CAT-CDIs differed in their settings as to when to finish the administration. Specifically, the Polish CAT-CDI was set so that the SE as low as 0.1 be reached (with as few items as possible), and if the SE could not be lowered to or below 0.1, the CAT-CDI stopped after administering a maximum of 75 items. The

English CAT-CDI on the other hand was set to administer a maximum of 50 items, but the administration could be stopped earlier if the SE of 0.15 or lower was reached. In other words, the Polish CAT-CDI prioritized a lower SE, while the English CAT-CDI prioritized shorter test length.

Item selection in the two CAT-CDIs

It is to be expected that a CAT-CDI will not need to administer all the items from the CDI item bank. In fact, in all the task administrations in the validation study, the English CAT-CDI used altogether 251 items (36.91%) and the Polish CAT-CDI used altogether 258 items (38.74%) from their respective item banks. By design, a CAT-CDI selects items that provide the most information for each participant. In practice, this means CAT draws from a subset of items that both match the participant's current ability estimate in difficulty and show high discrimination, allowing clear distinctions between individuals with similar ability levels. A potential concern, however, is that a CAT-CDI might repeatedly rely on the same subset of optimal items. CAT-CDI's usefulness in research—particularly in longitudinal studies—lies in the possibility of administering the CDI frequently without parents becoming overly familiar with specific, frequently repeated items. This advantage holds only if the CAT-CDI does not consistently present the same items across administrations. Therefore, we investigated whether the two CAT-CDIs repeatedly selected any items too frequently. In English, 3 items appeared in more than half of administrations - *"long"* in 64% of administrations, *"make"* in 54% of administrations, and *"last"* in 53% of administrations. As expected, they are also high discrimination items in their difficulty bins. "Long" is additionally one of the starting items, i.e. a first item shown to the parent, chosen so that there is high probability the parent can respond positively. In Polish, 3 items appeared in more than half of administrations - *"szukać"* (to look for) appearing in 83% of administrations, *"znaleźć"* (to find) appearing in 61% of administrations, and *"babcia"* (grandmother) appearing in 60%

of administrations. As expected, all three items show very high discrimination. *"Szukać"* (to look for) and *"znaleźć"* (to find) are in fact in top 5 discrimination items in general. *"Babcia"* (grandmother) is one of the starting items. The fact that only 3 items out of c.a. 250 (c.a. 1%) are used in more than half administrations is to CAT-CDI's advantage.

Since CAT tailors item selection to a child's ability estimate, parents of children at different ability levels may be expected to receive largely distinct sets of items. However, children with similar ability estimates may be administered items from the same difficulty range, which increases the likelihood of repeated exposure to a particular subset of items. This is not inherently problematic, but in the context of longitudinal testing with frequent administrations, it raises the possibility that parents may encounter very similar sets of items when the child's ability remains relatively stable.

Thus, we also examined whether parents of children with similar ability estimates were administered the same items in the CAT-CDI. To do so, we divided ability estimates into deciles and calculated the frequency of each item's occurrence within each theta decile (see Figure 3).

Item properties in the two CAT-CDIs

Our second aim was to analyze similarities and differences in IRT item properties in CAT in English and Polish.

There are 679 items in the English CAT-CDI and 666 items in the Polish CAT-CDI. In both languages, the items' difficulty and discrimination parameters were calculated using IRT 2 parameter model (these included separate samples, see Kachergis et al. 2022 and Krajewski et al., in preparation). An item's difficulty indicates the ability level at which there is a 50% probability that a participant will respond correctly (here: that a child will know a word). Difficulty is expressed on the same scale as theta (measured ability): negative values correspond to harder items, values near zero to items of moderate

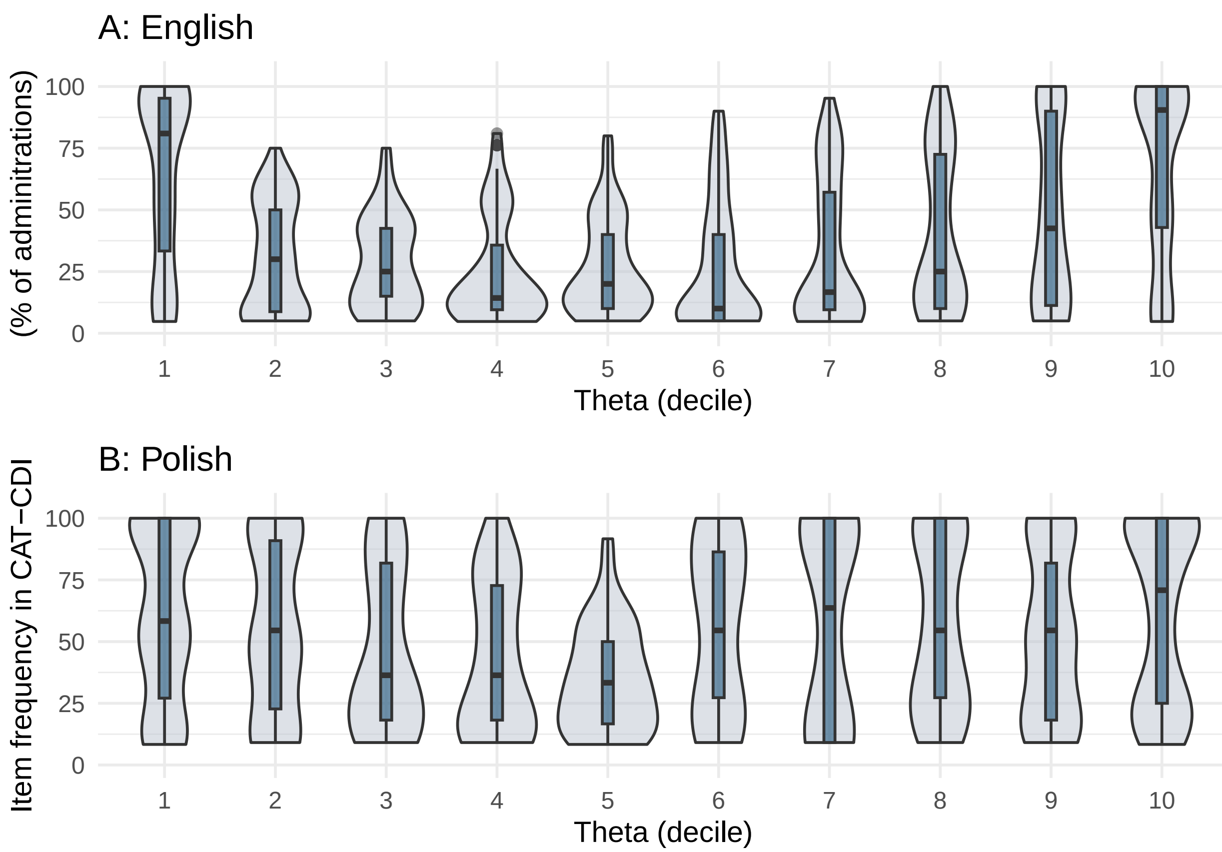


Figure 3

difficulty, and positive values to easier items (note that difficulty is inversely related to theta and so, negative theta means low ability while positive theta means high ability). An item's discrimination shows how well it can distinguish between individuals whose abilities are close. In this paper, we focus mainly on item difficulty, as it is directly tied to the ability being measured, whereas discrimination reflects more the quality of the item as a measurement tool.

English items were in general more difficult than the Polish items, $\Delta M = -1.87$, 95% CI $[-2.06, -1.69]$, $t(1227.61) = -20.09$, $p < .001$ (English: min = -7.16, max = 4.45, M = -2.19, Mdn = -2.21, SD = 1.98; Polish: min = -4.34, max = 4.41, M = -0.32, Mdn = -0.43, SD = 1.41). This was true also for a subset of 395 items common to both languages - English items still proved to be more difficult than the Polish ones: $M_D = -1.68$, 95% CI

$[-1.82, -1.53]$, $t(394) = -23.26$, $p < .001$. This is a result of the characteristics of the samples used to estimate the item parameters. In English, item difficulty was calculated based on a broader sample of children aged 12–36 months, spanning the CDI:WG, CDI:WS, and CDI-III (reflecting the idea of having one CAT-CDI for all the age versions), whereas the Polish data came from a sample of narrower age range of 18–36 months, corresponding to the CDI:WS. As a result, item difficulty in English was estimated using a relatively younger sample, for whom certain items may have been more challenging – thus appearing more difficult – compared to the older Polish sample. Still the item difficulty for common items in the two languages was positively and moderately correlated: $r = .65$, 95% CI $[.59, .71]$, $t(393) = 17.09$, $p < .001$.

We also examined whether CDI semantic categories show comparable difficulty values across the two languages. To this end, we conducted a series of Spearman's rank correlations on item difficulty within each semantic category (we considered only items common to both the Polish and English CDIs) (see Table 3). The rank correlations coefficients varied by category, but for half of the categories the correlations were moderate to strong (0.52 to 0.91).

The least correlated semantic category was Quantifiers. The discrepancy may be partially to the fact that many English quantifiers map onto multiple Polish equivalents - for example, "some" maps to two Polish CDI items, "jeszcze" (with a similar difficulty value as the English "some") and "więcej" (more difficult than "jeszcze" and "some"). These two Polish items are used in different contexts, i.e., "jeszcze" would be used more often in child directed speech than "więcej" also because it can be used to refer to both quantity and repetitive actions (meaning "again"). "Więcej" implies a quantitative comparison of some sorts, as in "more than before" or "more than two".

Interestingly, the second least correlated semantic category were Clothes. Here, some divergence may be explained by one-to-multiple mapping (e.g., "hat" is either "czapka" or "kapelusz" in Polish), but also words characteristics, e.g., "bib" in Polish is a much more complex word, "śliniaczek"; English "underpants" are more morphologically complex (and maybe less frequent) than Polish "majtki".

Taken together, these two semantic categories reflect the existing cross-linguistic differences in lexical mapping as well as item-specific factors such as frequency, morphological complexity, and contextual usage that .

Table 3

Cross-linguistic correlations of item difficulty within each CDI semantic category (Spearman's rank correlations).

CDI category	rho	n
quantifiers	0.11	10
connecting_words	0.20	9
clothing	0.22	19
locations	0.31	6
pronouns_demonstrative	0.32	4
places	0.32	7
vehicles	0.45	9
furniture_rooms	0.47	17
action_words	0.50	75
body_parts	0.52	17
descriptive_words	0.55	23
prepositions	0.55	10
games_routines	0.57	12
time_words	0.57	8
household	0.57	30
sounds	0.60	6
food_drink	0.63	41
outside	0.63	21
people	0.67	16
animals	0.69	30
toys	0.74	15
question_words	0.91	10

Last, we investigated whether particular lexical categories show related difficulty values across the two languages. We considered the following lexical classes: verbs, nouns, adjectives, function words and other, i.e. mostly sounds and words relating to routines (e.g. “hello”, “yes”, “thank you”) and time (e.g. “tommorow”, “day”). Again, we performed a series of Spearman’s rank correlations (on the difficulty of items common to CDI-CATs in Polish and English) for each lexical class (see Table 4. The weakest correlation was for function words (0.40), then verbs (0.47), adjectives (0.53), nouns (0.6), other (0.75). Notably, function words include many of the CDI semantic categories which showed the lowest correlations in Table 3, i.e., quantifiers, connecting words, locations, and pronouns. Again, many English function words map onto multiple Polish equivalents, e.g., “which” corresponds to “jaki” (when asking about type, kind, or quality) and “który” (when selecting from a limited set of options). These cross-linguistic differences partially explain the weaker correlation for function words. What is more, because function words are structurally tied to grammar (which is highly language-specific) they naturally vary in frequency in child-directed speech and in their communicative salience, which can affect how well children know them across languages. By contrast, the relatively high correlations for nouns, sounds and words relating to routines (grouped together as “other”) reflect their consistent easiness/difficulty across languages: words such as “mum”, “ball”, “hi”, “woof woof” are acquired early, aligning with findings that nouns are strongly and consistently over-represented in early vocabularies across many languages (e.g., Frank et al., 2021).

Table 4

lexical_class_en	rho	n
function_words	0.40	50
verbs	0.49	71
adjectives	0.53	25
nouns	0.59	194
other	0.75	50

Discussion

It seems then that the relation between the two estimates differs, depending on the ability level: for lower abilities, the difference between the two estimates is most visible, with CAT-CDI estimates higher than the full CDI estimtes. For higher ability, the difference decreases or even (as in the case of Polish) changes direction, with CAT-CDI estimates lower than the full CDI estimates.

In points for now:

1. *Strong correlations with full CDI.*
   The CAT-CDIs were strongly correlated with the full CDI versions in both languages, which supports the overall validity of the CAT approach. These correlations held for both monolingual and multilingual participants. However, the Polish sample was small, so further research is needed—and planned—to confirm these findings.

2. *Cases of extreme divergence.*
   CAT-CDI scores diverged notably from full CDI estimates for only a small number of children in both language groups. In the Polish sample all divergent cases had low SEM, and their full CDIs were unusually short–maybe the full form underestimated

their ability? In the English sample, their full CDIs were not very short, but SEM was relatively low, which is good.

3. *Item paramteres across languages.*
   Item parameters (i.e., difficulty and discrimination) showed moderate correlations across languages. These likely stem from variations in the calibration samples. Still, some semantic categories showed consistent patterns across languages, suggesting underlying structural similarities in lexical development.

4. *Item selection.*
   Each CAT version used c.a. half of the full CDI item pool. Few items were overused (mostly those with highest discrimination in the item pool or starting items). But we also saw items being chosen for CAT that wouldn't be considered very informative (i.e., of low discrimination). That's something to explore further, I wrote to Phil Chalmers (author of mirt and mirtCAT), maybe he'll be able to help?

5. *Efficiency and Reliability.*
   CAT-CDIs were significantly shorter than the full CDIs while maintaining high reliability for most children. The few less reliable estimates were for children at the ends of the ability range (children with either very low or very high lexical skills). Comparing Polish and English samples, the ability ranges with low SEM differed (see Figure 3) - this again might be a result of the calibration samples and item parameters? This is a good place to stress the importance of well-balanced, diverse calibration datasets. Also, while the CAT-CDI is efficient and useful for screening or initial identification ("pomiar przesiewowy" in Polish), it may be less suited for diagnostic purposes where high precision is needed across the full ability range.

References

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